Control Variates for a GARCH Model

Zero-Variance MCMC and Lasso-Based Control Variate Selection

LELOUP Baptiste MAGHREBI Mohamed Aziz BERLIA Alexandre

ENSAE Paris

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Motivation

The problem

Posterior expectations $\mu_f = \mathbb{E}_\pi[f(\omega)]$ are estimated by MCMC averages:

$$\widehat{\mu}_{\mathrm{MC}} = \frac{1}{N}\sum_{i=1}^{N} f(\omega^{(i)}).$$

Unbiased but high variance: halving the error requires more iterations.

Our approach (Mira et al., 2013)

Replace f by a corrected function $\widetilde{f}$ such that

$$\mathbb{E}_\pi[\widetilde{f}] = \mathbb{E}_\pi[f] \quad \text{but} \quad \mathrm{Var}_\pi(\widetilde{f}) \ll \mathrm{Var}_\pi(f),$$

as a *post-processing step* on the same chain, no extra MCMC cost.

Outline

Q1 MCMC baseline for the Bayesian GARCH posterior

Q2 First-order Zero-Variance control variates

Q3 Second-order controls: OLS vs Lasso vs Post-Lasso

Q4 Validity of regression on autocorrelated chains

Conclusion

Each question follows the structure: theory → implementation → results.

The Bayesian GARCH(1,1) Model

Model. For log-returns $r_t = \log(S_t/S_{t-1})$:

$$r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, h_t), \qquad h_t = \omega_1 + \omega_2\, r_{t-1}^2 + \omega_3\, h_{t-1}.$$

Parameter constraints (positivity & stationarity):

$$\omega_1 > 0, \quad \omega_2, \omega_3 \geq 0, \quad \omega_2 + \omega_3 < 1.$$

Prior. Truncated normals on $\mathbb{R}_+$: $\omega_k \sim \mathcal{N}_{\mathbb{R}_+}(0, 100)$, weakly informative. The last constraint $\omega_2 + \omega_3 < 1$ is enforced directly in the log-posterior evaluation.

Targets. $\mathbb{E}[\omega_1 \mid r]$, $\mathbb{E}[\omega_2 \mid r]$, $\mathbb{E}[\omega_3 \mid r]$, $\mathbb{E}[\omega_2 + \omega_3 \mid r]$, the last is the *persistence of volatility*.

Log-Posterior Derivation

By Bayes' rule: $\log \pi(\boldsymbol{\omega} \mid r) = \log L(r \mid \boldsymbol{\omega}) + \log \pi_0(\boldsymbol{\omega}) + c$.

Log-likelihood (Markov factorisation):

$$\log L(r \mid \boldsymbol{\omega}) = -\tfrac{1}{2} \sum_{t=1}^{T} \left[\log(2\pi) + \log h_t + \frac{r_t^2}{h_t} \right].$$

Final log-posterior (up to an additive constant):

$$\boxed{\log \pi(\boldsymbol{\omega} \mid r) = -\tfrac{1}{2} \sum_{t=1}^{T} \left[\log h_t + \frac{r_t^2}{h_t} \right] - \tfrac{1}{2} \sum_{k=1}^{3} \frac{\omega_k^2}{s_k^2} + \text{const.}}$$

- Returns $-\infty$ if admissibility constraints are violated.
- The unknown normalising constant cancels in MCMC ratios and in log-posterior gradients.

Q1 - Random Walk Metropolis-Hastings

Algorithm

At iteration k, propose $\boldsymbol{\omega}^\star = \boldsymbol{\omega}^{(k)} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \Sigma_{\text{prop}})$. Accept with probability:

$$\alpha = \min\left(1,\ \frac{\pi(\boldsymbol{\omega}^\star \mid r)}{\pi(\boldsymbol{\omega}^{(k)} \mid r)}\right).$$

The unknown normalising constant cancels in the ratio.

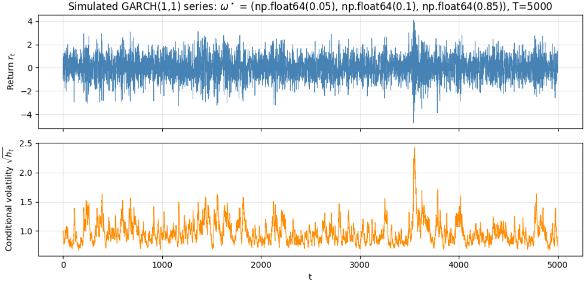
Adaptive tuning (Roberts-Gelman-Gilks)

Pilot run $\to$ empirical covariance $\widehat{\Sigma}_\pi \to \Sigma_{\text{prop}} = (2.38^2/d)\,\widehat{\Sigma}_\pi$.

Why a baseline matters. ZV estimators are computed from posterior draws generated by the baseline RWM sampler. They reduce estimator variance but do not fix poor mixing of the underlying chain.

Q1 - Simulated GARCH Series

Setting. $\omega^\star = (0.05, 0.10, 0.85)$, $T = 5{,}000$.



The simulated series exhibits the expected GARCH behaviour: returns are centred at zero, but the conditional volatility $\sqrt{h_t}$ varies over time, with periods of high volatility followed by high volatility (*volatility clustering*).

Q1 - Posterior Diagnostics on Simulated Data

Reference chain ($N = 300{,}000$):

$$\widehat{\mathbb{E}}[\omega \mid r] = (0.067, 0.097, 0.835)$$

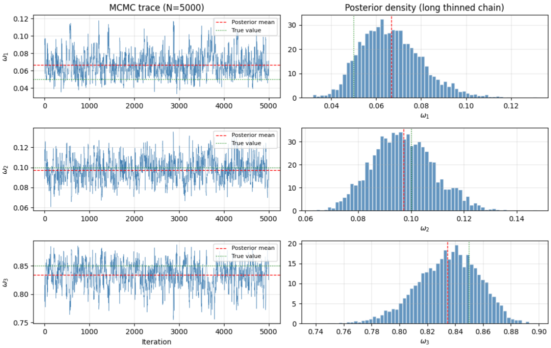
$$\widehat{\mathrm{sd}}(\omega \mid r) = (0.014, 0.012, 0.022)$$

$$\mathrm{ESS} \approx (14070, 15340, 12290)$$

Acceptance rate: 0.40.

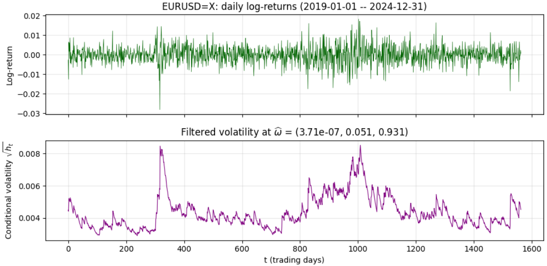
Persistence: $\widehat{\omega_2 + \omega_3} = 0.93$.

- Posterior means close to truth.
- ESS values confirm good mixing.
- Stationarity respected.



Left: short-chain traces. Right: posterior densities from the thinned N=300,000 reference chain.

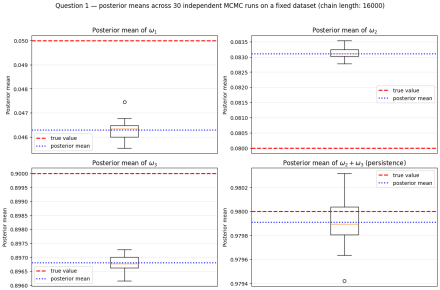
Q1 - Application to EUR/USD Log-Returns



- Daily log-returns from 2019 to 2024, demeaned.
- First pilot acceptance rate 0.023 (too low) $\rightarrow$ re-tuned with `adaptive_rwm_fx` $\rightarrow$ acceptance 0.31.
- Posterior estimates: $\widehat{\boldsymbol{\omega}} = (3.7 \times 10^{-7},\ 0.051,\ 0.931)$, persistence $\widehat{\omega_2 + \omega_3} = 0.982$.
- **Lesson**: ZV is post-processing, if the chain mixes badly, ZV breaks.

Q1 - Quantifying the Monte Carlo Error

Experimental design. Fix one simulated dataset; run $n_{\text{rep}} = 30$ *independent* chains varying only the MCMC seed. The spread across runs = pure Monte Carlo error, the quantity ZV is designed to reduce.



The box width is the **baseline MC error** we will compare against in Q2 and Q3.

Q2 - The Zero-Variance Principle

Central formula (Mira et al., 2013, eq. 8):

$$\boxed{\widetilde{f}(\omega) = f(\omega) - \tfrac{1}{2}\Delta P(\omega) + \nabla P(\omega)^\top z(\omega)}, \quad z(\omega) := -\tfrac{1}{2}\nabla \log \pi(\omega \mid r).$$

Two key properties

$\mathbb{E}_\pi[\widetilde{f}] = \mathbb{E}_\pi[f]$ (unchanged target), and $\mathrm{Var}_\pi(\widetilde{f}) \ll \mathrm{Var}_\pi(f)$ for a well-chosen P.

Degree-1 polynomial: $P(\omega) = \boldsymbol{a}^\top \omega$

$\nabla P = \boldsymbol{a}$, $\Delta P = 0$, so $\widetilde{f}(\omega) = f(\omega) + \boldsymbol{a}^\top z(\omega)$.

Q2 - The Link with Linear Regression

Optimal coefficient. Minimising $\mathrm{Var}_\pi(\widetilde{f})$ gives:

$$\boldsymbol{a}^\star = -\Sigma_{zz}^{-1}\,\mathrm{Cov}_\pi(z, f), \qquad \Sigma_{zz} = \mathbb{E}_\pi[zz^\top].$$

These expectations are not closed-form, but estimated by ergodic averages:

$$\widehat{\Sigma}_{zz} = \tfrac{1}{N}\sum_i z_i z_i^\top, \qquad \widehat{\mathrm{Cov}}(z, f) = \tfrac{1}{N}\sum_i z_i\,(f_i - \bar{f}).$$

The cornerstone

$\hat{\boldsymbol{a}} = -\widehat{\Sigma}_{zz}^{-1}\widehat{\mathrm{Cov}}(z, f)$ is exactly the OLS coefficient from regressing $(f(\boldsymbol{\omega}^{(i)}))_i$ on $(z(\boldsymbol{\omega}^{(i)}))_i$ along the chain.

Implementation in 5 steps. chain $\rightarrow$ compute z $\rightarrow$ build matrix W (3 columns) $\rightarrow$ OLS regression $\rightarrow$ $\hat{\mu}_{\mathrm{ZV}} = \bar{f} - \hat{\boldsymbol{a}}^\top \bar{W}$.

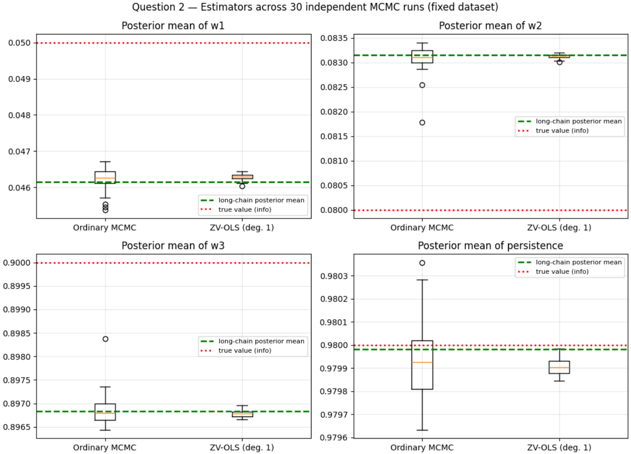
Q2 - First-Order Results

$n_{\text{rep}} = 30$ independent chains on a fixed dataset. Compare ordinary MCMC vs ZV-OLS using the 3-column degree-1 design.

Variance reduction factors

Target	VR factor
ω_1	13.5
ω_2	49.9
ω_3	24.8
$\omega_2 + \omega_3$	18.6

- 10–50$\times$ reduction.
- No visible bias.
- Negligible cost (one OLS).



Q3 - Second-Order Controls

Degree-2 polynomial: $P(\omega) = \boldsymbol{a}^\top \omega + \frac{1}{2}\omega^\top B\omega$

$$\widetilde{f}(\omega) = f(\omega) - \tfrac{1}{2}\mathrm{tr}(B) + (\boldsymbol{a} + B\omega)^\top z(\omega).$$

For $d = 3$: canonical basis with $\frac{1}{2}d(d+3) = 9$ regressors.

Why enrich the control space?

Degree-1 captures only the *linear* component of f explained by the score. Degree-2 enriches the control space with terms involving $\omega_i z_j$, capturing more of the posterior geometry and often yielding much stronger variance reduction

New challenge. With more regressors, the OLS design may become collinear or unstable, motivates regularised alternatives.

Q3 - Three Estimators Compared

ZV-OLS

Full OLS on 9 regressors, variance-optimal in i.i.d. theory, but unstable when controls are collinear or when the dictionary becomes large.

ZV-Lasso CV

$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta}\left[\frac{1}{2n}\|f - W\beta\|_2^2 + \lambda\|\beta\|_1\right], \quad \lambda \text{ chosen by CV.}$$

Selects relevant controls (sets some coefficients to zero) but *shrinks* the surviving ones (introduces bias).

ZV-Post-Lasso

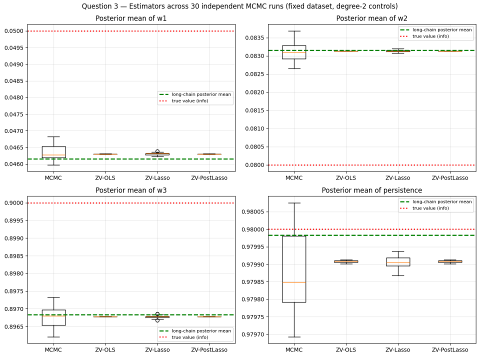
Use Lasso only for variable selection, then refit OLS on the selected support $\hat{S} = \{j : \hat{\beta}_j^{\text{lasso}} \neq 0\}$. Combines selection with no shrinkage.

Q3 - Numerical Results on Simulated Data

Target	ZV-OLS	ZV-Lasso	ZV-Post-Lasso
ω_1	1,612	39.9	1,612
ω_2	10,394	103.8	10,394
ω_3	6,274	49.0	6,274
$\omega_2 + \omega_3$	1,389	50.2	1,389

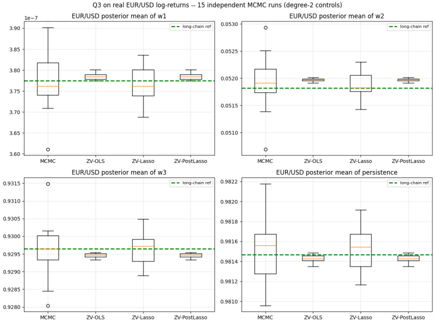
- Degree-2 brings **2-3 extra orders of magnitude** vs degree-1.
- Plain Lasso *underperforms*: shrinkage is the wrong default when all 9 controls genuinely contribute.
- Post-Lasso $\equiv$ OLS, Lasso simply selects all 9 controls, so refitting recovers the OLS solution.

Q3 - Visualising the Variance Reduction



- ZV-OLS and ZV-Post-Lasso boxes are **visually invisible** next to MCMC, variance shrunk by 3-4 orders of magnitude.
- ZV-Lasso boxes still much narrower than MCMC, but wider than OLS.
- All boxes centred on the long-chain reference.

Q3 - Application to Real EUR/USD Data



Real-data performance

Variance reduction on EUR/USD (degree 2, ZV-OLS): 80$\times$ to 350$\times$.

This is smaller than on simulated data because:

- the FX posterior is much more concentrated, so absolute MC noise is already small;
- ω_1 sits close to the boundary $\omega_1 > 0$, making integration-by-parts boundary terms non-negligible.

Q4 (Bonus) - Validity of OLS on an MCMC Chain

The concern. OLS theory assumes i.i.d. samples. MCMC states are correlated: does this break the regression step?

Two natural remedies

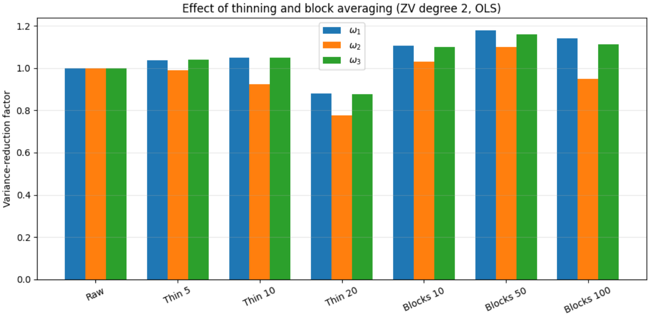
Thinning ($k = 5, 10, 20$): keep one sample every k to fit $\hat{\beta}$ on near-decorrelated states.

Block averaging ($B = 10, 50, 100$): regress on means of consecutive blocks (nearly i.i.d. for $B \gg \tau$, the integrated autocorrelation time).

Trade-off

Both methods reduce the *effective sample size* for the regression (N/k or N/B instead of N). The benefit must outweigh this cost.

Q4 - Empirical Verdict



- Raw chain is competitive and thinning/block averaging brings no robust improvement.
- Thinning *degrades* performance for large k (less data for $\hat{\beta}$).
- Block averaging brings no benefit (N/B effective samples).

Conclusion

- Zero-Variance control variates proved to be a powerful post-processing tool for improving MCMC estimation in the Bayesian GARCH(1,1) model.
- Degree-1 controls already reduce Monte Carlo noise substantially, while degree-2 controls exploit the posterior geometry much more efficiently.
- In this low-dimensional setting, OLS remains the most effective choice; Lasso becomes more relevant when the control dictionary is larger or more collinear.
- The method remains effective on real FX data, provided that the underlying MCMC sampler is well calibrated.
- Empirically assessed autocorrelation effects: in this setting, using the raw chain was as good as thinning or block averaging.

Thank you for your attention

Questions?

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